# Shivanshu Siyanwal

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shivanshu-siyanwal    Uttar Pradesh, India


## PROFESSIONAL SUMMARY

Quantum Software Engineer with 4+ years of experience in quantum algorithm development, production-grade Python software, and hybrid quantum-classical workflows. Proficient in Qiskit, Cirq, PennyLane, and cloud platforms (AWS Braket). Demonstrated expertise in quantum error correction, variational quantum algorithms (VQA/VQE/VQLS), and CI/CD for simulation infrastructure. Published researcher in QML, QKD, and photonic quantum computing. Seeking to develop and optimise scalable quantum software for real-world devices.

## TECHNICAL SKILLS

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|---|---|
| **Quantum Frameworks** | Qiskit, Cirq, PennyLane, Perceval, Amazon Braket, CUDA-Q, QTensorFlow |
| **Quantum Domains** | Quantum error correction, VQAs, linear optical QC, quantum network simulation, tenso |
| **Software Engineering** | Python (5+ years, production code), C++, Git, CI/CD pipelines, Linux |
| **ML/AI** | PyTorch, TensorFlow, NumPy, SciPy, Scikit-learn |
| **Cloud & Backend** | AWS Braket, FPGA deployment (AMD Alveo), HLS, HBM, REST APIs |
| **Simulation Tools** | OptiSystem, ANSYS, QuTip, QETLAB |

## WORK EXPERIENCE

### CDAC (Centre for Development of Advanced Computing) — Noida, India

*Quantum Applications Engineer* — Oct 2024 – Present

- Developed hybrid quantum-classical pipelines for weather forecasting using VQC (7 qubits, 50 parameters), achieving 56.5% improvement over statistical baselines; production code deployed on cloud simulation infrastructure.
- Implemented and optimised quantum algorithms (VQE, QCNN, fidelity-based kernels) with Qiskit and PennyLane; built scalable software modules for quantum image encoding (FRQI, QPIE, NEQR).
- Deployed custom VQC kernel on AMD Alveo U55C FPGA using HLS and HBM, achieving 6.7× speedup over GPUs (291.8 s inference); integrated into CI/CD pipelines for continuous integration and testing.
- Collaborated with cross-functional teams (hardware, research, engineering) to optimise hybrid workflows and support cloud-based quantum computing platforms.

### Dr. Arpita Maitra, IAI, TCG CREST — Kolkata, India

*Project Junior Research Fellow – Quantum Networks & Linear Optics* — Mar 2024 – Sep 2024

- Designed and simulated linear optical quantum network testbeds (DPS, COW, Twin-Field QKD) using OptiSystem and ANSYS, modelling realistic photonic channel impairments, loss, and noise.
- Performed error modelling and security analysis against intercept-resend and PNS attacks; contributed to resource state generation and entanglement generation benchmarking.
- Built Python post-processing tools to manage optical simulation data, enabling reusable and maintainable software for quantum network simulation.

### Dr. Arpita Maitra, IAI, TCG CREST — Kolkata, India

*Research Intern – Quantum Cryptography & Circuit Design* — Aug 2023 – Mar 2024

- Designed quantum circuits for block ciphers (Mini-AES, S-AES) and Galois Field arithmetic; evaluated on Qiskit simulators and IBM Quantum hardware.
- Demonstrated chosen-ciphertext attack using Grover's search and amplitude amplification; contributed to error correction and circuit optimisation.

### Arqanum Technologies Pvt. Ltd. — Pune, India

*Junior Quantum Computing Developer* — Sep 2022 – Jul 2023

- Implemented and optimised VQC, QBM, and VQLS for time-series forecasting and financial analysis; used AWS Braket and Rigetti hardware for hybrid quantum-classical linear system solving.
- Developed quantum data encoding techniques (amplitude, phase, angle) and Ising Hamiltonian circuits; contributed to MeitY-QCAL project on Dicke state simulation.

## RESEARCH EXPERIENCE

**NMR QC Group, Prof. Kavita Dorai – IISER Mohali** **Punjab, India**

*MS Research Student – NMR Entanglement Detection & Quantum Simulation* Aug 2021 – Aug 2022

- Designed neural network models for genuine multipartite entanglement detection and SLOCC classification in 2- and 3-qubit NMR systems; published in *Quantum Information Processing* (Springer).
- Performed experimental density matrix reconstruction and quantum error behaviour modelling; applied PCA, MCA, LDA for quantum dataset analysis.

## EDUCATION

**Indian Institute of Science Education and Research (IISER) Mohali** **Punjab, India**

*Five-Year Dual BS-MS (Postgraduate) – Major: Physics; Minor: Data Sciences* Aug 2017 – May 2022

**Dayawati Modi Academy (DMA-I)** **Uttar Pradesh, India**

*High School Diploma (CBSE) – 95.8%* May 2014 – May 2017

## SELECTED PUBLICATIONS & CONFERENCES

- **[IEEE TQCEBT 2026]** Hybrid Variational Quantum Circuits based Temperature Forecasting with Low-Latency FPGA Inference (accepted).
- **[IEEE AI-ML Bangalore 2025]** Quantum Machine Learning Models for Enhanced Weather Prediction.
- **[Springer Nature – Under Review]** Quantum Image Representations based Quantum Neural Networks for Binary Classification.
- **[Quantum Information Processing, Springer]** Entanglement Detection in 3-Qubit NMR States via ANNs (published).
- QTML 2024 (Univ. of Melbourne) – Poster: Quantum Techniques in Machine Learning.
- QCC2025 (IAI, TCG CREST) – Poster: R&D on Quantum Communication & Computation.

## CERTIFICATIONS & AWARDS

- IBM Certified Associate Developer – Quantum Computation using Qiskit v0.2X (2026)
- Qiskit Global Summer School 2022 – Quantum Simulations (Advanced Level)
- Qiskit Global Summer School 2021 – Quantum Machine Learning (Certificate of Excellence)
- IBM Quantum Challenge Africa 2021 – Advanced Achievement
- INSPIRE Fellow (SHE) – DST, Ministry of Science & Technology, Govt of India (2017–2022)
- TOEFL iBT – Score 96 (2022)